AWASIR91

PEDIATRICS

كبريت & عميري

.....



بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

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أنت الذي ما فتئت تقلب وجهك شطر الأحلام مقدماً من السعي قرباناً وترجو أن يحتويك قدراً ترضاه

قد شهد الله والله لا يضيع سعي الساعين

ف اطمئن

PART 1: NEONATOLOGY

1. Important terms
2. Maternal diseases
3. Drugs
4. Congenital infections
5. Birth injuries
6. Neonatal care
7. New born resuscitation
8. Neonatal seizures
9. Respiratory distress in neonates
10. Other problems of prematurity
11. Neonatal infections
12. Neonatal jaundice
13. Primitive reflexes

1. Important terms:

- **Embryo:** < 9 weeks
- **Foetus:** 9 weeks - birth
- **Still birth:** born ≥ 24 weeks with no signs of life
- **Prenatal mortality** = Still birth + up to 1 week of delivery/ 1000 X live still births
- **Neonatal mortality** = Death in first 4 weeks of delivery/ 1000 X live births

- **Pre-term:** < 37 weeks
- **Term:** 37 - 41 weeks
- **Post-term:** > 42 weeks
- **Low birth weight:** < 2500g
- **VLBW:** < 1500g
- **Extremely LBW:** < 1000g
- **Small for GA:** BW < 10th centile
- **Large for GA:** BW > 90th centile

- ✓ **Neonate:** up to 28 days
- ✓ **Infant:** 1 - 12 months
- ✓ **Toddler:** 1 - 3 years
- ✓ **Pre School:** 3 - 5 years
- ✓ **School:** 5 - 12 years

إذا فعلت اليوم ما يتكاسل غيرك عن فعله ،

فسوف تفعل غداً ما يعجز غيرك عن القيام به.

2. Maternal diseases:

- Most diseases affect foetus and neonate are:

1- DM. 2- Graves. 3- SLE.

4- AITP (auto immune thrombothytopenic purpura).

1) Problems related to diabetic mother:

1- **Congenital malformations**:

- Most common: Neural tube defects and cardiac malformations.
- Characteristics:

a- Caudal regression syndrome "sacral agenesis"

b- Hypo plastic left colon

- Control of DM decrease risk

N.B. Congenital malformation **does not** increase in Gestational DM

2- **IUGR**

3- **Macrosomia:**

- Shoulder dystocia
- Cephalo-pelvic disproportion
- Birth injuries

In neonate, increase insulin can cause:

1- Hypoglycaemia > (early feeding + close monitoring), Hypocalcaemia, Hypomagnesaemia, Hyperbilirubinemia.

2- RDS (Insulin Anti cortisol)

3- Polycythaemia (Neonatal jaundice) -

4- Transient HoCM (hypertrophic obstructive cardiomyopathy) (resolves in few weeks but may cause heart failure)

2) Graves:

- Antibodies cross the placenta
- Foetus> Tachycardia = CTG, goitre = US
- Neonates > irritability, weight loss, and tachycardia = Heart failure.... Diarrhoea
- Xerophthalmous
- May require Antithyroid

3) SLE

- Anti Ro/ Anti La
- Neonatal lupus syndrome (Self limiting rash +/- heart block

4) AITP:

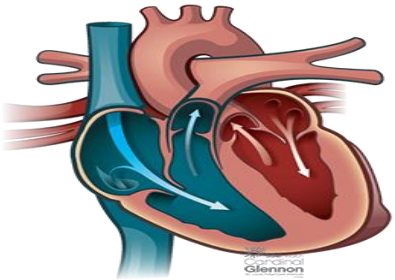
- Low platelets = increase intracranial haemorrhage
- If at delivery = petichea/severe Thrombocytopenia = I.V IGs
- If active bleeding = platelets.

3. Drugs:

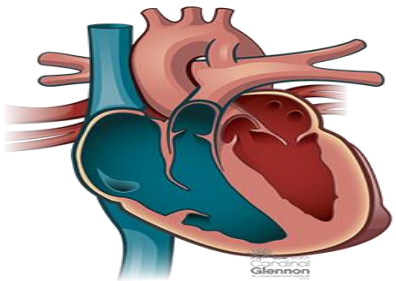
Drug	Effect
1- Warfarin	1. Cartilage (nasal Hypoplasia/Epiphyseal stippling) 2. Haemorrhage
2- Lithium	Ebstein's Anomaly
3- Phenytoin	Hydantoid face (mid facial Hypoplasia)
4- Thalidomide	Phocomelia (short limbs)
5- Tetracycline	Enamel Hypoplasia
6- Diethylstilbestrol (DES)	Vaginal/CX adenocarcinoma
7- Alcohol	1-fetal alcohol syndrome 2-cardiac abnormalities
8- IV drugs	Withdrawal (jitteriness, seizures, yawing)
9- Smoking	stillbirth/miscarriage , LBW, IUGR

Ebstein anomaly:

Rare heart defect in which the tricuspid valve doesn't work properly, as a result blood leaks back through the valve into the right atrium.



Healthy Heart



Ebstein's Anomaly

Foetal alcohol syndrome:

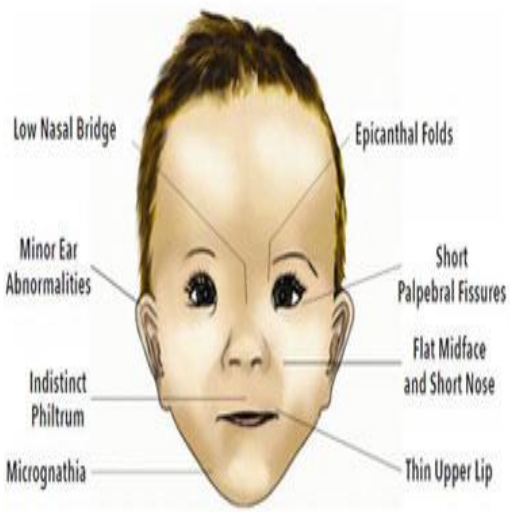
1- **Characteristic face**

- Flat nasal bridge
- Maxillary Hypoplasia
- Absent philtrum
- Short thin upper lip

2- **Growth abnormalities**

3- **Developmental delay**

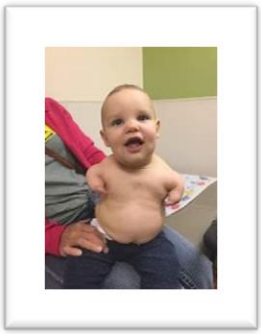
4- **Cardiac abnormalities**



PHENYTOIN - Fetal Hydantoin syndrome

- Cleft lip/palate
- Microcephaly
- Mental retardation





Phocomelia

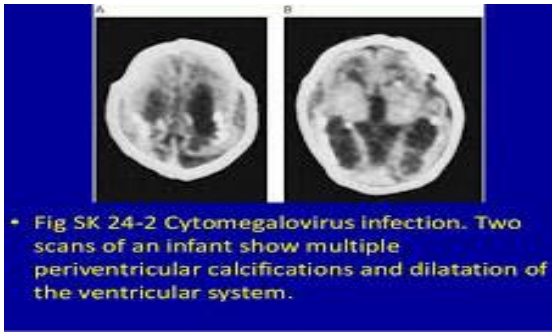
4. Congenital infections

Infection	c/f	Dx	Tx
1- **CMV**	Most common 90% Asymptomatic 5% Early symptoms Periventricular calcification. - Others: Hepatosplenomegaly, SN deafness, Microcephaly. Primary infection is the most serious	urine PCR for virus/culture IgG antibodies show previous infection.	Gancyclovir, Sedofovir, Foscarnet
2- rubella	- Earlier > Most dangerous - Triad of: 1- Cataract 2- SN deafness	IgM antibody detection	

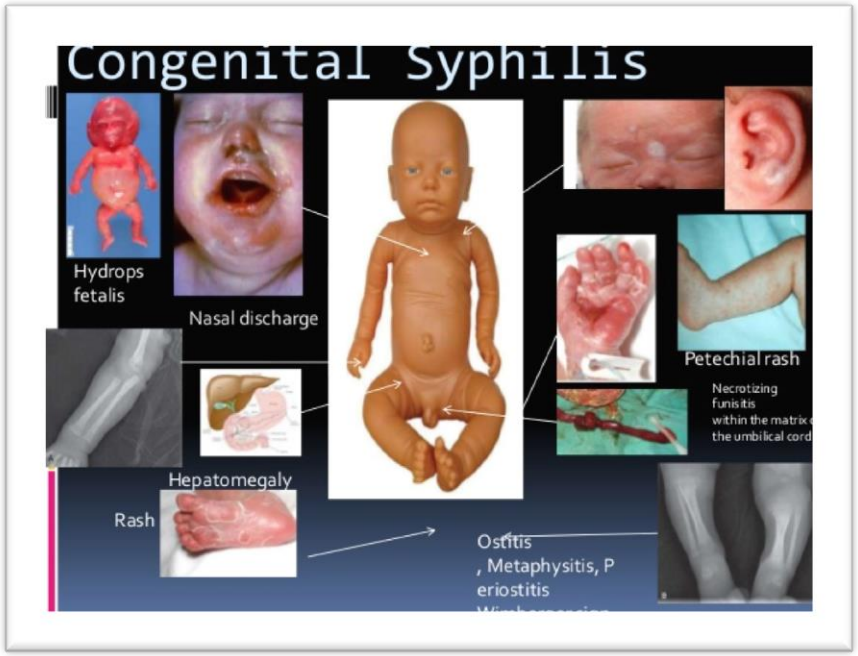
	3- CHD (e.g. PDA) - Others: Blueberry muffins spots, extra medullary haematopoiesis - After 4 months of gestation > No infection		
3-Toxoplasma	- Triad of: 1- Diffuse calcification 2- Chorioretinitis 3- Hydrocephalus > Long term neurological disability	IgM or IgG (later)	Pyramethamine + Sulfadiazine for 1 year
4-varicella	<20 wks ~Fetopathic (Embryopathic varicella) - Zigzag scarring of skin, Limb atrophy, digital dysplasia. - Mortality 30% - Rash, pnuomonitis, hepatitis. - Very severe disease		VZ IG + IV acyclovir for the mother
5-Syphilis	≥ 2 years ~ Hutchinson's triad 1- H. teeth 2- interstitial keratitis 3- 8th SN deafness - Other: rash on palm / sole (early), bone deformity (saber shin and saddle nose)		Penicillin (mother 1 month or more before delivery)

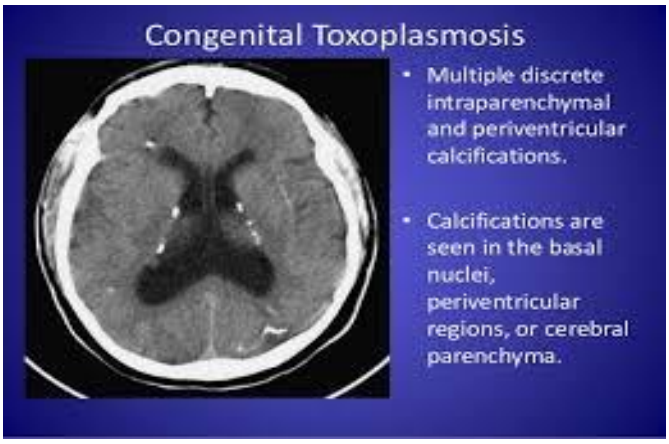


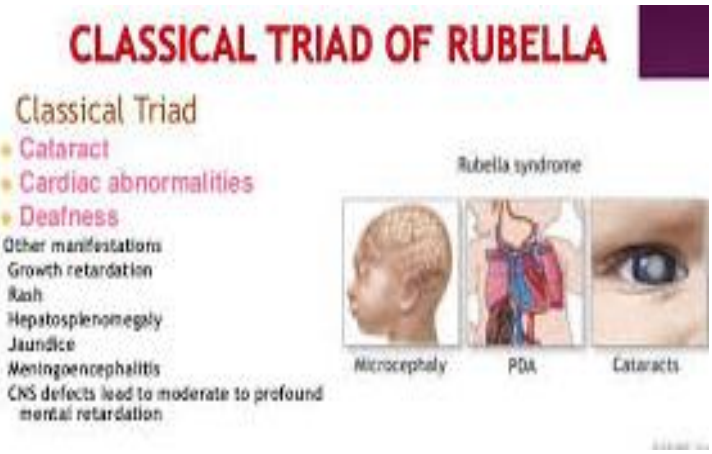




- Fig SK 24-2 Cytomegalovirus infection. Two scans of an infant show multiple periventricular calcifications and dilatation of the ventricular system.







5. Birth injuries

Soft tissues:

1- Caput succedaneum:

- Oedema (diffuse) - abnormal skin over it - resolves in few days.

2- Cephalohematoma

3- Subperiosteal bleeding (usually involve parietal bone)

4- Subaponeurotic haemorrhage – shock, coagulopathy

Nerve injuries: (shoulder dystocia, breech)

1- Brachial plexus:

*** Erb's:** - confined (do not cross sutures).

- association: skull fractures #, I.C. haemorrhage.

- complications: neonatal jaundice Resolves in few weeks

2- Chignon: ventuse delivery

- In upper root of BP C5, C6

- Waiter's tip deformity

- Palm +ve grasp reflex

- Phrenic nerve palsy.

*** Klumpek's:**

- In lower root of BP C8, T1

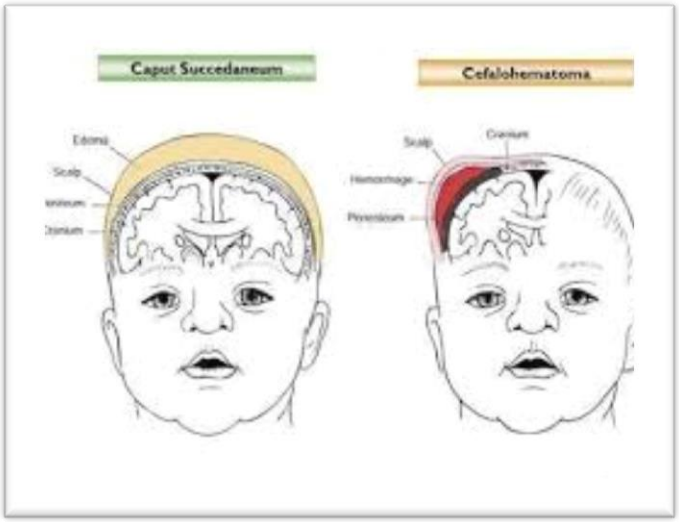
- Claw Hand

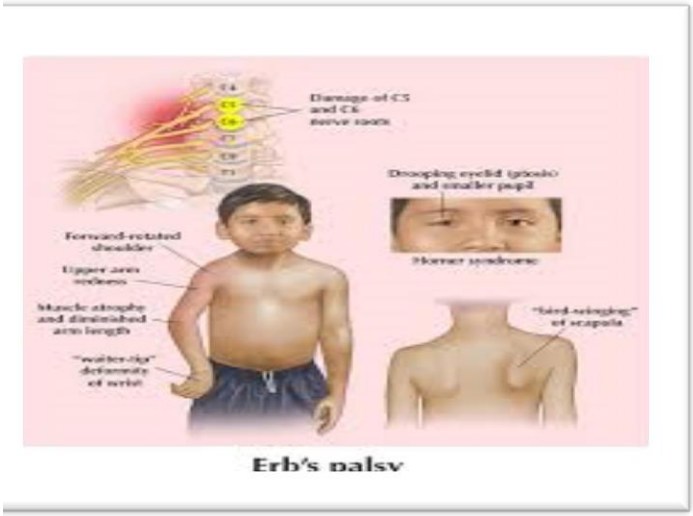
- Small muscle wasting in hand

- -ve grasp reflex - Herne’s

3- Facial nerve injury: (forceps) Need methyl cellulose to protect the eyes

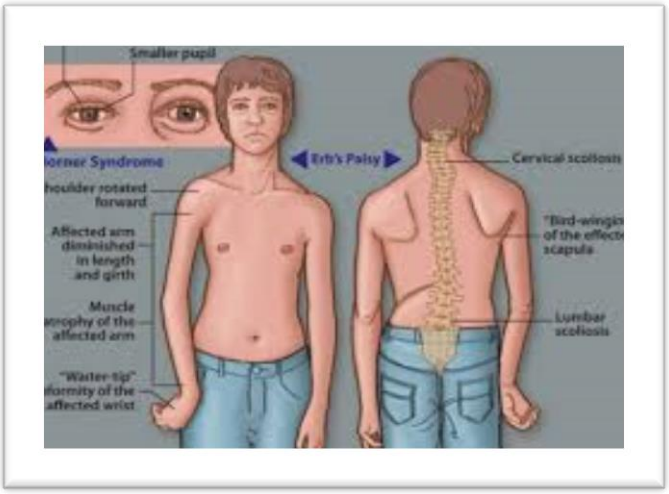
#Bones: Most common is the clavicle



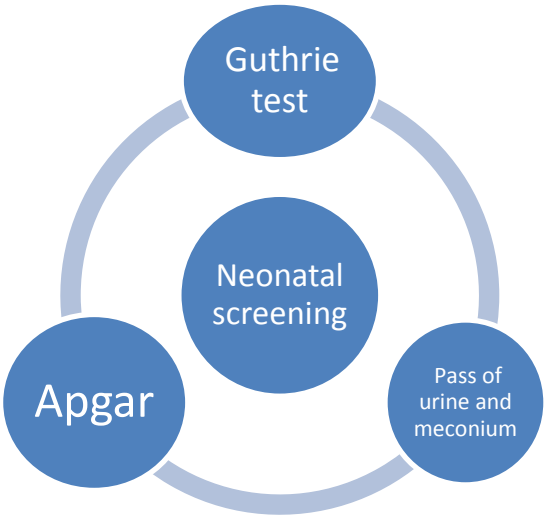


Erb's palsy





6. Neonatal care



1- Erythromycin eye drops

Prevention of Gonococcal/Clamydial conjunctivitis

- Silver nitrate > against Gonococcalonly ..causes chemical conjunctivitis

2- Wash with Hexachlorophene

(in preterm > neurotoxic)

3- Vitamin K

4- Vaccination

5-Umblical cord care

6- Hearing test

2- if not > renal agenesis or hypovolemia

Passage of meconium:

- 80% in 1st 24h

- 99% in 48h

- If not > check for intestinal obstruction or ano rectal anomalies

Passage of urine:

- 25% in 1st 24h

- If not check the bladder:

1- if palpable > obstruction

Apgar Scoring System

Indicator		0 Points	1 Point	2 Points
A	Activity (muscle tone)	Absent	Flexed arms and legs	Active
P	Pulse	Absent	Below 100 bpm	Over 100 bpm
G	Grimace (reflex irritability)	Floppy	Minimal response to stimulation	Prompt response to stimulation
A	Appearance (skin color)	Blue; pale	Pink body, Blue extremities	Pink
R	Respiration	Absent	Slow and irregular	Vigorous cry

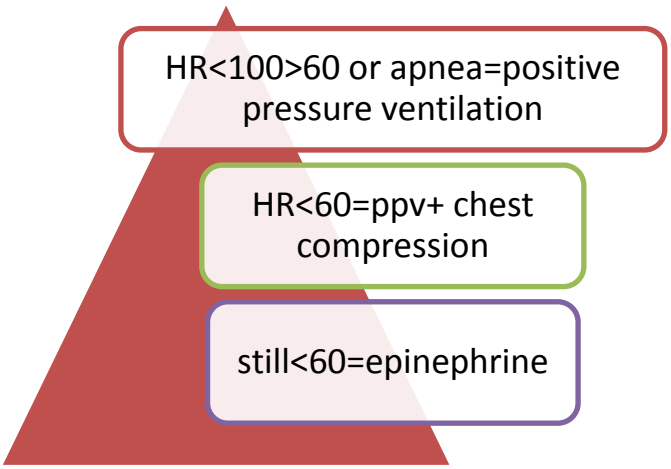
Normal = 8 - 9 (1min - 5 min) # 4 - 7 (attention)# 0 - 3 (immediate resuscitation)

7. New born resuscitation:

▪ Observe for:

- Meconium
- Term
- Good muscle tone
- Good cry
- Pink colour
- If Yes >> Routine care
- If No >> Provide warmth

Clear airway + O2 100%, check HR, Respiration.



- **N.B:** most common cause => respiratory
- Aim of chest compression => provide blood to the heart not the brain like adults
- Chest compression = PPV 3:1
- In 1 minute: 90 chest compression: 30 breaths
- Site: lower 1/3 of the sternum just below an imaginary line between the nipples. Thumb or two fingers method (less effective but easier)
- Diameter depress to 1/3 of anterior-posterior of the chest.
- Cause of respiratory depression = opioids by mother
- Give Naloxone (0.1 mg/kg/dose) "if RD continues despite resuscitation"

Hypoglycaemia: Glucose<45 mg/dL (2.6 mmol/L) need for neurological action (primary fuel for the brain).

Risk 1. Mother DM 2.Premature (<37 wk) i.e., glycogen in foetus is formed in the last month 3. SGA decreased stores / IUGR 4. LGA increased demands. 5. Illness / hypothermia increased demands. 6. Certain drugs. **Complain of** neurological symptoms + sympathetic Sx(tremors, tachycardia, tachypnea)
Treatment Oral feeding If comatose: IV 10% D (2 ml/khg) If no response: increase the rate or the concentration central venues catheter to avoid skin necrosis. If you can't give 10% D or still no response gives glucagon + Hydrocortisone.
Whom to treat

1- Symptomatic infants 2- Asymptomatic but 2 abnormal readings 3- Asymptomatic but 1 very low reading <1.6 mol/dL

Hypothermia:

- Normal temp : 36.5 - 37.5°C (axillary)
- 36 – 36.4 : cold stressors
- 32 -35.9° : moderate hypothermia.
- < 32° : severe hypothermia.

Or by touch:

- Cold trunk + cold extremities = severe hypothermia.
- Warm trunk + cold extremities = cold stress

Risk factors:

1. Pre maturity (low brown fat) like SGA
2. Neurological problem, cardiac, respiratory no control
3. Large pores in the body (e.g. spina bifida)
4. Sedation.

N.B: Thermo-regulation in infants:

- Brown fat (barring). Brown fat doesn't form until late.
- Hypothalamic control to neither release nor epinephrine => vasoconstriction in periphery & pulmonary. No shivering.

Treatment: Dryness, heater, O2 + glucose (burning; release heat).

N.B: losing of heat:

1. Convection: air flow.
2. Conduction: direct contact.
3. Radiation: no direct contact.
4. Evaporation.
5. **HIE (hypoxic ischemic encephalopathy):**

Abnormal neurological behaviour caused by hypoxic ischemic brain injury due to asphyxia.

Is the single most important cause of prenatal mortality and morbidity

Causes

- Ante partum /peri partum / post partum
- ***Placenta:*** Abruption, excessive contraction, ruptured uterus
- ***Umbilical cord:*** Prolapsed or compression
- ***Blood:*** hypo or hypertension of the mother
- ***Fetal:*** e.g. IUGR
- ***Neonatal:*** kerinctrus, inborn error of metabolism

Classification of HIE

Feature	G1mild	G2 moderate	G3 severe
Consciousness	irritable	Lethargic	comatose
Tone	Hypotonic	Marked	sever
seizures	No	Yes	prolonged
suckling	Poor	unable(tube feed)	unable to maintain resp.+/-MOF

Diagnosis: There is no clear diagnostic test for HIE

- Abnormal findings on the neurologic exam in the first few days after birth, is the single most useful predictor that brain insult has occurred in the perinatal period

Essential criteria for diagnosis of HIE:

- Metabolic acidosis (cord pH <7 or base deficit of ≥12
- Early onset of encephalopathy
- Multisystem organ dysfunction

Characteristic MRI finding. Predominant pattern in preterm: *Periventricular leukomalacia (spastic diplegia)*

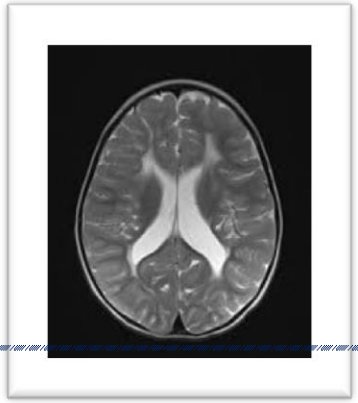
In term babies: *Parasagital cerebral injury (proximal quadraparesis*)

Sequel:

- CP
- Neuronal damage: primary (immediate), secondary (respiratory failure)

Management:

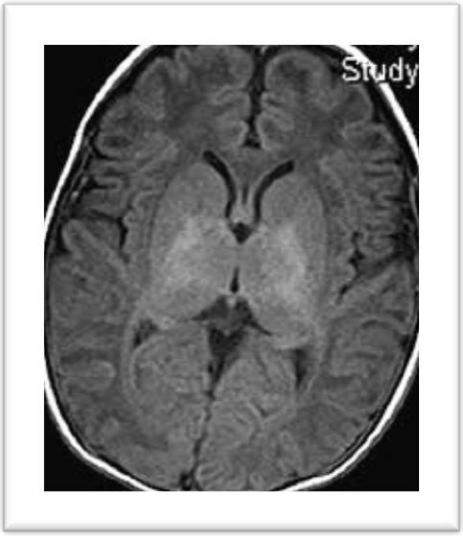
- Supportive care, ABCD approach
- Prevent further brain damage



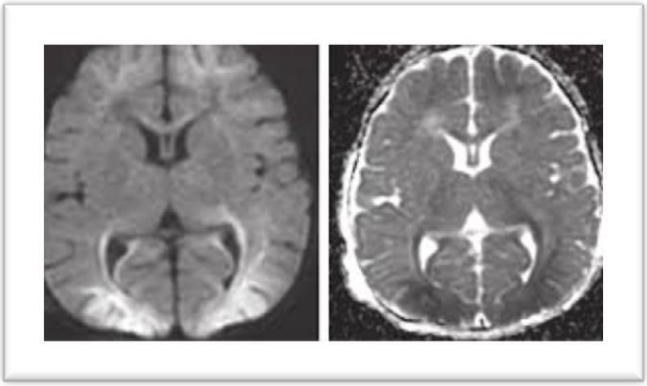
Periventricular leukomalacia

في الأصل..الأوقات كلها صباحات

دعها فقط تشرق من داخلك



Neonatal HIE



Parasagital brain injury

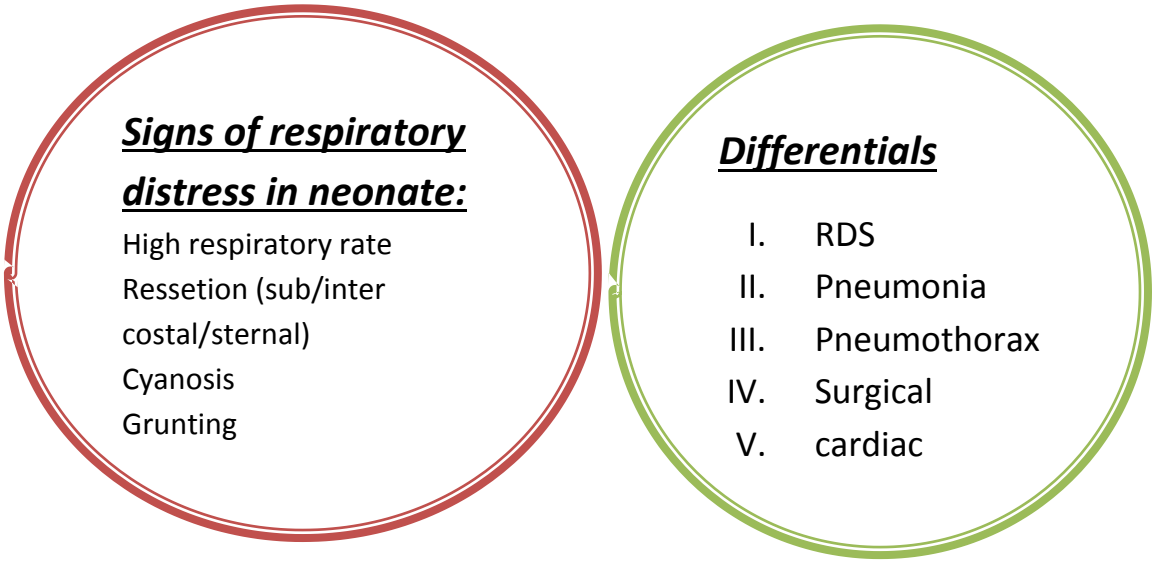


Child with HIE

8. Neonatal seizures:

Etiology	Work up	Treatment
At birth: maternal anaesthesia injected into foetus **Day 1:** 1. Metabolic; hypoglycaemia/ hypocalcaemia 2. HIE => 6- 18h become more severe in the next 24-48h. 3. Birth trauma => 12h or more. primary SAH, cerebral venues thrombosis 4. CNS intrauterine infections & sepsis. 5- Drug withdrawal/ L.A. toxicity **24-72hrs:** 1- Cerebraldysgenesis. 2- Vascular (infracts, haemorrhage). 3- Intracranialhemorrhage. 4- birth trauma +confusion 5- Tubulousclerosis. **72hrs-1 week** 1-. Cerebral malformation/infracts 2-. Familial neonatal seizures 3-. Hypoparathyroidism 4-. Kerinctrus	**First line:** 1. Pulse oximetry. 2. Glucose. 3. PCV. 4. Serum Ca. (Ionized),Na, Mg. 5. Arterial blood gases. 6. LP, blood cultures. 7. Cranial USS. 8. EEG. ***Second line:*** 1. MRI or CT. 2. Maternal & neonatal samples for drugs. 3. Virology/Congenital infection screen. 4. Serum ammonia & amino-acids. 5. Urine amino & organic acids. 6. Therapeutic trial of pyridoxine	***- Treat underlying cause*** ***-O2 then give glucose*** * seizure prophylaxis ***First line***: - phenobarbitone 20-40 mg/kg then 5mg/kg ***2nd line***: phenytoin 10-20 mg/kg ***3rd line***: diazepam 0.3 mg/kg * prognosis depends on the cause

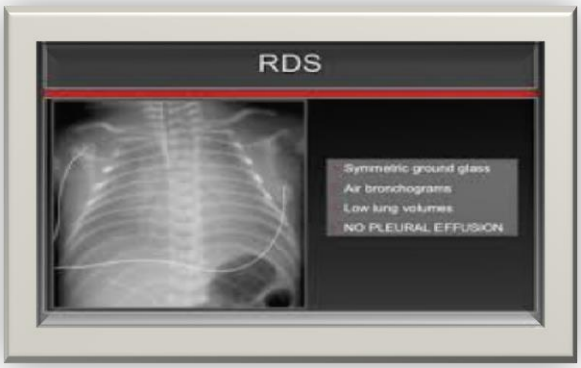
9. Respiratory distress in neonates:



** IUGR is protective i.e. stress → ↑cortisol

RDS: Surfactant deficiency.

- Surfactant is produced by type 2 alveolar cell late ≥ 32 weeks.

Risk factors
-prematurity Type 1 DM -male
Diagnosis
--Confirmatory: CXR - bilateral diffuse ground glass appearance of lung - airbronchogram - atelectasis 
Treatment
-O2 (best initial tx) - Exo.surfactant by ET (best effective tx) - Respiratory support (CPAP Or IPPV - Fluids/Abx - Monitor by → vital signs, ABG
Complications
1. **PDA** (HF+murmur+collapsing pulse) 2. **Air leak** (Pneumothorax, Pneumomediastainum, Emphysema) 3. **chronic lung disease** 4**. Cerebral palsy**
Prevention
prevent premature delivery , if already occur→ IV **steroids** for 48 hours →Betamethasone (12 mg/24 hours X 2 doses) OR dexamethasone (4 mg/12 hours X 4 doses)

Management of PDA: - Restriction of fluids + diuretics

- Indomethicin (medical) - Surgery

Management of chronic lung disease

•Ventilation with low PO2 and high CO2 via tracheotomy by CPAP. Mechanical ventilation if indicated only

•steroids
•give RSV prophylaxis
-improve lung function: bronchodilator, fluid restriction, diuretics, and steroids

10. Other problems of prematurity:

****Apnea of prematurity** (immature resp center+ low HR)

DX of exclusion

(Apnea +high HR = subtle seizures)

Apnea: no breathing for ≥ 20 sec

TX: gentle stimulation, Xanthenes tx, CPAP if recurrent

****Retinopathy of prematurity:**

- Due to oxygen therapy, damages growing retinal vessels>> blindness

- Laser therapy is sometimes needed, but it usually resolves alone.

****** **Jaundice of prematurity:** - Premature liver - Self limiting

****NEC (necrotizing enterocolitis):**

- Cow's milk is a risk factor - Toxic child: distended shinny abdomen, central apnea, bloody diarrhea, bilious vomiting

-Dx: Abdominal X ray

- Pneumonaleintestinalis

- Air in portal tract

-Air under diaphragm if perforated

- distended bowel loops

- Tx :

No entral feeding for 10 days (NPO, IV fluids, NGT)

IV broad spectrum Abx

If perforation: surgery

- Complications:

Perforation>peritonitis, sepsis

Stricture> long term

Short bowel syndrome>malabsorption

Diaphragmatic hernia:

- Left sided in 85%

-resp. distress, failure to resuscitation (heart displacement to the right)

- Dx: CXR→ bowel in chest+ heart displacement

- Tx: first step →NGT suction to decompress stomach

Then surgical repair

11. Neonatal infections:

Early < 72h (pneumonia and meningitis)	**Late** > 72h
1- Group B strep (GBS) 2- E.coli 3- Listeria	- Nasocomial - S.epidermidis - If pneumonia (S.aureus, P.arginosa)
~ Septic screen: 1- CBC > Neutropenia 2- CRP 3- Blood culture 4- Urine culture 5- CXR 6- LP 7- Throat, tracheal, skin, rectal swab	
~ Mx: Benzyl penicillin (GBS) +Gentamycin (E.coli) + Ampicillin (Listeria) Or ampicillin + gentamycin	

- Prolong rupture of membranes > 18 h
  Chorioamnionitis (fever, tender abdomen, foul smelling discharge)
- Preterm +ve GBS, in high vaginal swab
- Why preterm has increase risk?? IgG from mother cross placenta in the last month of 3rd trimester.
- If mother increase R.F > take high vaginal swab
  +ve GBS > Intrapartum Benzylpenicillin
  Late onset > Vancomycin + Gentamycin

Meningitis	***Conjunctivitis***	***Meconium aspiration syndrome***	**Transient tachypnea of the newborn: TTN**
- High mortality - ***ttt:*** Penicillin / Ampicilin+ 3rd generation cephalosporin (14 - 21 day) - ***Sequele***: abscess, hydrocephalus, hearing loss, Neurodevelopmental problems * Ceftriaxone displace Bilirubin > Jaundice **** N.B*** sx not specific (eg; no stiffness) > Bulging of fontanlle, Lepithalamus :late ▪CXR for pneumonia = RDS	▪ ***first 48 h***: Sticky eyes > cleaning by water or saline Discharge + redness > Staph or Strept - ***ttt:*** topical antibiotics - Purulent discharge + injection of conjuctiva + swellen eyeled Gonococcal (most serious) ➡ - ***ttt:*** I.V 3rd generation cephalosporin - ***Dx***: gram stain, culture ▪ ***7 - 14 days:*** - Chlamydia (most common) - ***ttt:*** Oral erythromycin for 2 weeks - ***Dx:*** immunoflurocent stain	Meconium swolled with amniotic fluid Post term, breech, fetal hypoxia ***Effect:*** - Collapse > acute airway obs. (complete/partial) - Chemical pneumonitis - RDS - Air leak ~ ↑ 2ry pneumonia - Persistent pulmonary hypertension of newborn - ***CXR***: hyper inflation/ flattened diaphragm	bening self limiting RDS of mature (&CS) #1 / late premature - delayed clearance of lung liquids - appears shortly after birth - resolve in 3 - 5 days ▪ Clearance of lung by Increase trans pul pressure ***R.F:*** 1- C.S 2- Male 3- Breech 4- Precipitus delivery (extremly rapid labor) ***C/O:*** - increase RR - respiratory distress - increase AP of chest (barrel chest) ***On examination:*** - crackles - palpable liver and spleen ***Investigations:*** ***CXR***: Gold standard > Hyperinflation (Perihilar streaking , flat diaphragm) No consolidation Dx of exclusion ***MX:*** O2 / tmp. Exclude sepsis Feeding (avoid in RR >60/min ie;aspiration ***Prevention:*** - CS at 39weeks - ante natal steroid before CS

*** DDx of increase ***RR*** in newborn: (TRACHEA)
- TTN
- Respiratory infection
- Aspiration
- Congenital malformation
- Hyaline membrane disease (RDS)
- Edema
- Air lead. / Acidosis

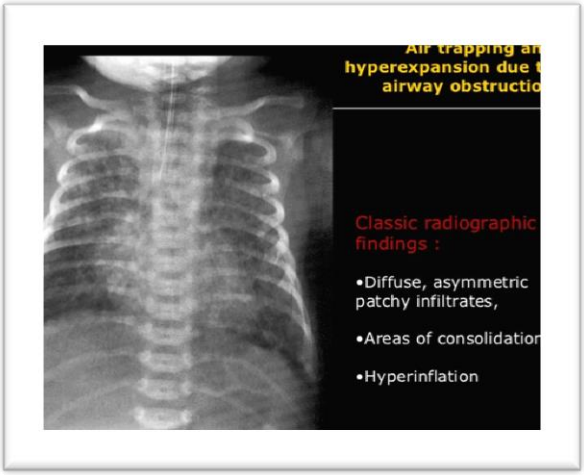
other causes of **Persistent pulmonary hypertension** of newborn

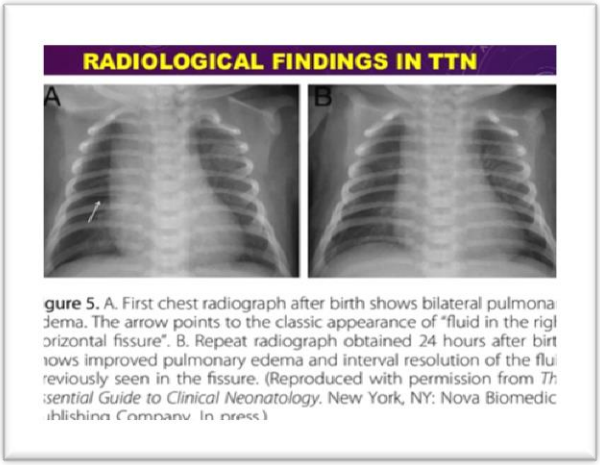
Birth asphexia, septicemia, RDS or a 1ry disorder

Cyanosis, HF , murmur are often absent-

ttt: mechanical ventilation + circulatory support

Vasodilators: inhaled NO , Sildenfi (viagra)
if very severe: extra corpreal membrane oxygenation in ECMO





ستصافح بيديك النجوم يوماً، ستصبح قمراً مشعاً ....

فقاوم لطفا ....قاوم

12. Neonatal jaundice:

Causes:

▪ **1st day:**	- Hemolysis: ABO incompatibility, G6PD, HS,Congenital infection (conjugated) - Blood group Rhesus: serious, 2nd child, Hepatosplenomegaly, Hydrops anemia ABO : Commoner, 1est child may be affected, NO hepatosplenomedgaly Rhesus: mother Rh -ve, fetus Rh +ve ABO: mother O, fetus usually A Dx: by coombs test
From **2nd day:**	- Physiological - Breast milk jaundice - B. feeding jaundice - Infection (UTI)

☆Physiological jaundice:
Dx of exclusion
1-NEVER on day 1
2- Peak 3rd/4th day (later in premature)
3- Not more than 12mg/dl
4- Early increase not more than 5mg/dl
5- Disappears in 7th day
Causes:
- Liver immaturity.
- decrease life span (80 days)
- increase RBC mass (hypoxic enviroment

Breast milk jaundice:
Breast milk contain inhibitors of conjugation
- increase enterohepatic circulation
- ttt: stop B.F for 1 - 2 days

VS
Breast feeding jaundice -→ ttt; fed the infant

Jaundice:Face (5mg/dl) > neck > upper chest (10mg/dl) > abd (12 mg/dl) > palm / soles (15mg/dl)

Management:

● If reach 16 mg/dl (lower in premature):-
→ Phototherapy (blue/white light)

● If 20 mg/dl :-
>> Exchange transfusion > twice blood volume (2 X 85 X kg >> through umblical Venous catheter

Side effects:

Phototheapy	***Exchange transfusion***
- lethargy , retinal damage - Nasal abs. by eye pads - Mask cyanosis - Maculopapular rash - Increase insensible loss > dehydration - Diarrhoea > dehydration	Blood → Infection, transfusion reaction, met. Catheter → Perforation, emboli Procedure→ hypotension, NEC, porto splenic thrombosis

☆Kernicterous: Bilirubin encephalopathy:

- Unconjugated bilirubin > lipid soluble > crosses BBB
- Early > poor feeding/lethargy
- If severe → high pitch cry, Increase tone (opisthotonos), Convulsions, coma
- Deposition:In Basal ganglia/ hippocampus & Brain stem nuclei

☆ Polycythemia:

- HCT: > 65% - ttt: Portal exchange transfusion (replace blood by saline)
- Amount: blood volume (85) X HCT - desired HCT / observed HCT **Desired HCT = 50%

▪ ***Un conjugated***	- ***Conjugated:*** (never physiological)
1- Breast milk 2- Infection (blood, urine) 3- Hypothyroidism in TFTs 4- Pyloric stenosis	1- Neonatal hepatitis (viral, metabolic e.g. cystic fibrosis, alfa 1 antitrypsin def. `most common genetic cause`) 2- TORCH > sepsis, hepatocellular damage 3- Biliary atresia/ Choledocal cyst (U/S + biopsy) ... ttt: surgery (kassi operation > Porto entrostomy)

- Sequels:
Choreoatheliod CP



SN deafness
Mental retardation
Enamel dysplasia

Keep smile

13. Neonatal reflexes:

▪ ***Moro***: up to 4 months
- How: sudden drop of head/noise
>> Extension and abduction of upper limbs followed by flexion and adduction of Lower limbs

- If absent bilaterally	***- If absent unilaterally***
- CNS damage (HIE, Kernictrues, haemorrhage) -Bilateral Erb's -Sepsis	-Clavicle fracture - Shoulder dislocation - Erb's - Fracture humerous -Septic arthritis

***** SN deafness in neonate:***
- Kernictrues
- Rubella
- CMV
- Syphilis

هذا العالم يبدأ منك حين تنهض بهمة . وينطفئ تماما حين تشعره بأنه لا فائدة منك..

Part 2: MALNUTRITION

1. Introduction
2. Marasmus
3. Kwashicor
4. IDA
5. Management of malnutrition
6. Rickets

Types of Nutrients

Type 1 (Copper,Ca2+,I-,Fe2+ and vitamins)	**Type 2 (essential aminoacids , Na+,K+ الجدول الدوري)**
Functional	Growth
+ve body stores	Nobody stores
Signs of deficiency	No signs of deficiency
↓ tissue concentration with deficiency	No ↓ in tissue concentration with deficiency (stable con.)
Variable in breast milk	Stable in breast milk
Growth failure is not a feature	Growth failure

Classification of PEM (protein energy malnutrition)**:**

*Welcome Classification:

Class	**Weight (% of standard forage)**	**Oedema**
Underweight	60-80 %	No
Kwashiorkor	60-80 %	Yes
Marasmus	<60 %	No
Mar. Kwash	<60 %	Yes

Advantage= Easy

Disadvantage= No acute/chronic, age-dependent

NB: Standard weight=50th percentile

NB: age in 1st year approx. $=\frac{Age\ \text{"months"}+9}{2}$

In 2nd year= add 2.5kg

After 2 years= (Age "years" x2) +8

Difference between wasting and stunting:

	Weight/age	Height/age	Weight/height
Wasting	- or ↓	-	<5th percentile
Stunting	↓	↓	-

*** Water low classification**= - Wt/Ht. - Not Age dependent

*** Gomez**= Researchers/ field work (weight/age)

*** WHO**= Age-dependent < _ 2 Z score :> malnutrition

- Low weight/age= under nutrition

- Low weight/height= wasting= acute malnutrition

- Low height/age= stunting= chronic malnutrition

Marasmus

- Time: Early infancy
- Diet: Deficiency in both proteins and carbohydrates

Causes:

- Primary (Nutritional): Failure of BF, feeding difficulties (e.g. CP, prematurity, cleft palate)

- Secondary: Diarrhoea, malignancy, malabsorption, chronic disease etc.)

NB: Premature= feeding difficulty + rapid growth

Disease= Anorexia + ↑ catabolism + food intolerance

=Low (↓glycogen + diabetic pattern)

C/O:

1) **Growth failure**: Severe wasting, (↓weight/height), first failure to gain weight then height then H.C
2) **Loss of subcutaneous fat:** First abdomen, then limbs, buttock, last: buccal pad of fat. Gives wrinkled skin and old man face
3) **Muscle wasting + Hypotonic**: Scaphoid abdomen, stick like limb,
4) **Psychological**: Irritability, crying, little sleep

Head is large, eyes starring and there are mild hair changes **BUT** No dermatosis, no liver infiltration.

Lab:

-Proteins= (serum enzymes) = Normal

-Low urea (no exo.proteins)

-Glucose=Low (↓glycogen + diabetic pattern)

Kwashiorkor:

- Time: Weaning or post weaning (2-3years)
- Diet: ↓ protein, normal carbohydrates

Causes: usually infections

1) Inadequate diet (↓ protein)
2) No synthesis of protein (liver diseases)
3) Loss of proteins (nephrotic syndrome, burns, haemorrhage etc.)
4) Malaria/ helminthic infections
5) Other causes

Features:

Constant features	***Usual features***	***Occasional features***
1-**Growth failure** 2-**Oedema** (most constant). Mild→G. oedema (due to activation of RAA sys. → due to hyovolemia, hypoprotinemia) 3-**Disturbed fat/muscle ration**: Fat is relatively present, muscle hypotonia and wasting 4-**Psychological**: Apathetic, lacking interest (due to aromatic AA deficiency	1-**Hair changes:** Sparse (temple/occipital), thin, despigmented, easily pluck able=loss of sulphur containing amino acids (cysteine, methionine, pamtothenic acid) and defect in melanin and low pamtothenic acid) 2-**GIT manifestations:** Anorexia, Diarrhoea (lack of enzymes, infection, villous atrophy	1-**Dermatosis** (pathognomenic) : hyper/hypo pigmentation , desquamation, ulceration, petichea due to low Zinc, niacin, vitamin A, essential amino acids, adrenal gland abnormalities. Weeping dermatitis→ resemble bruising 2-**Hepatomegaly**: Fatty infiltration. ↑free FA synthesis,↑mobilization,↓oxidation,↓excretion

NB: Lack of dissaccridone= Fermentative diarrhoea (peri-anal excoriation)
Malabsorption of lipids= Defect in bile salt conjugation

Lab: 2 constant :-

- ↓Proteins
- Fatty infiltration of liver

-↓ Proteins (including enzymes except liver enzymes ↑): only ↑ in gamma globulin
…..Reversed albumin: globulin ration
-↓ Urea
-↓ Carbohydrates + Diabetic pattern (insulin deficiency and insensitivity)
-↓ Lipids
-↑ Na+, ↓K+, ↓ Total Mg2+(normal serum)

Poor prognostic features:

1) Hypoglycaemia, hypoprotinemia, hypoalbuminemia
2) Sepsis
3) Liver size >6cm, jaundice, purpura
4) Vitamin A deficiency

Complications:

- Diarrhoea → dehydration, electrolytes dist.
- Infections (More in Kwash. , serious is Bronchopneumonia)
- Hypothermia (More in Mars.)
- Hypoglycaemia
- ✓ Mars. → Bleeding (purpura)
- ✓ Kwash. → Heart failure

Deficiencies:

B1 (Thiamine)	Beriberi (HF, neuropathy)
B2 (Riboflavin)	chelosis, angular stomatitis, glossitis
B3	Pellagra (Dermatitis<neck, face>, diarrhoea, dementia, death)
B6	Peripheral neuropathy
B12	Megaloblasticanaemia
Vit D	Rickets
Vit C	Scurvy (bleeding gums)
Zinc	Dermatitis (acrodermatitisenteropathica)

IDA (iron deficiency anaemia)

- Cow's milk "folic"=goat
- Affects: Cognition/ behaviour (may be irreversible if < 2 years)
- ↑Febrile convulsions, ↓ immunity

Vitamin A deficiency:

WHO classification of Vit A deficiency

class	Ocular manifestation
XN	Night blindness
X1A	Conjunctival xerosis
X1B	Bitot's spots
X2	Corneal xerosis
X3A	Corneal ulceration <1/3 of cornea
X3B	Corneal ulceration >1/3 of cornea
XS	Corneal scarring
XF	Xerophthalmic fundus

***Extra ocular**:*

- ↑ infections(esp; Measles)
- metaplasia
- UT= hematuria/pyuria
- Follicular hyperkeratosis
- Also ↑ I.C.P

- ***Treatment schedule (orally):***
  - <6 months: 50,000 IU Vit A (yellow pill)
  - 6-12 months: 100,000 IU (blue pill)
  - >12 months: 200,000 IU (red pill)
- The dose is repeated in the next day, and after at least 2 weeks (total of 3 doses). ***Prophylaxis***= From 6 months, every 6months till 5 years

Dx: Plasma retinol, conj. Examination, vaginal scraping

NB: ↑ vitamin A= Teratogenicity/ Pseudotumourorcerebri

Management of malnutrition:

-MUAC (medium upper arm circumference) only valid from 6 months to 5 years.
Green(normal): >12,5cm
Yellow (MAM moderate acute malnutrition): 11-12.5cm
Red (SAM sever acute malnutrition): <11cm
*- 3SD: Severe wasting

BOX 1. DIAGNOSTIC CRITERIA FOR SAM IN CHILDREN AGED 6–60 MONTHS

Indicator	Measure	Cut-off
Severe wasting (*2*)	Weight-for-height (*1*)	< -3 SD
Severe wasting (*2*)	MUAC	< 115 mm
Bilateral oedema (*3*)	Clinical sign	

Management of SAM:

If with:1) Medical complication (e.g. fever, pneumonia) 2)Grade 3 (+++) oedema (general edema)

Next step is inpatient management.

If: 1)No medical complication 2) Grade 1 (dorsum of foot) or 2 (thigh)

Next step is appetite test by **RUTF(ready to use therapeutic food): no** water, no home contamination

* Pass= eat >2/3, is an out patient * Fail: eat <2/3, treated as an in patient

Out pt Mx OF SAM: By RUTF. After 6-8 weeks will change to MAM

Mx of MAM: Supplementary feeding program (SFP) : 1- Food based 2- Non-food based

IN Pt Mx of SAM: (10)

1. Treat/prevent hypoglycemia
2. Treat/prevent hypothermia
3. Treat/prevent dehydration
4. Correct electrolyte imbalance
5. Treat/prevent infection
6. Correct micronutrient deficiencies
7. Start cautious feeding
8. Achieve catch-up growth
9. Provide sensory stimulation and emotional support
10. Prepare for follow-up after recovery

1-Tx/prevention of hypoglycemia	**2- Tx/prevention of hypothermia:**
What is **hypoglycemia**?? B.S < 54mg/dl = 3mmol/L **Tx :** - 1- If the child is conscious ,50ml 10 m% glucose or sucrose oral/NGT - If uncon. → IV 10% dextrose 5ml/kg - Then F.75 every ½ hour for 2 hours 1- Abx **Monitor:** 1- Blood glucose , if < 3 repeat dose 2- Rectal temp ,if <35.5 , repeat dextrose 3- LOC , i repeat dextrose	What is **hypothermia??** Axillary temp <35 C rectal temp < 35.5 c **Tx:** 1-Rewarm the child (kangaroo technique, heater, lamp (net fluorescent) NOT hot bottle 2- feed straightway .F 75 3- Abx **Monitor:** 1- Temp (first every ½ hour if heater , if no heater used every 2 hours) 2- Blood glucose. **PREVENTION :** 1- Feeding 2hriy day + night : hypoglycemia ,hypothermia 2- Keep baby clothed. 3- Keep baby dry 4- Avoid exposure (unnecessary ex.)
N.B :hypoglycemia often exists with hypothermia (both are signs of infection)	

Tx / prevent dehydration:

One indication for I.V fluids in malnutrition?? SHOCK

Fluid of dehydration in malnutrition: RESOMAL

Why not ORS?? In ORS, Na is high and K is low

	Na	K
ORS	90	20
RESOMAL	45	40

How to prepare ReSoMal:

1L O.R.S packet, 2L water, 50 g sugar, 40 ml electrolyte (K) / mineral solution (mg).

How to give ReSoMal:

-5-10 ml/kg / 1 hr for 4-10hrs (alternating with F-75) oral or NGT

SIGNS OF DEHYDRATION:

Reliable	Not reliable
Hx of diarrhea	Lethargy
-thirst	Sunken eye
Recent sunken eye	Skin turgor
Weak /absent radial pulse	Dry membrane
absent urine flow	

Monitor??Every ½ hour for 2hrs then hourly for 6-12hr

1-**PR** (low) 2-**RR** (low) 3- **U.O.P** (high)

If child develop increase RR, increase PR, puffy eyes + edema = stop fluids + reassess in 1 hour

To prevent after ReSoMal Tx:

After each loose stool: give 50-100 ml of ReSoMal

N.B: any child with watery diarrhea, assume dehydration.

How to treat septic shock??

- I.V fluids :

* 15 ml /kg / hr

Options; 5% D, 5% D+ ½ N.S, 5% D + R.L (ringer lactate)

- Broad spectrum Abx

N.B: achild is TX by those fluids but not improving or show signs of OHF what to do??

Slow blood transfusion 10 ml/kg over 3 hrs

4- Correct electrolyte imbalance:

Malnutrition: high Na, low K, low Mg

So we give extra k, Mg (electrolyte mineral solution)

How much?? 20 ml in 1L of food

5- Tx /prevent infection:

Usual signs of infection are not found (e.g. fever), so any child with malnutrition:

1 - **Abx:**

-Amoxicillin 15mg /kg 8hrly (no complication, oral) OR

-Ampicillin + gentamicin 50mg /kg 8hrly I.V /I.M (sever ill).

2- **Measles vaccine** if> 6months and not given.

Some add metronidazole 7.5mg /kg 8hrly

- If child is HIV +ve ,add cotimxazole

- If infection is identified →specific Tx.

- If no improvement in 48 hrs, add chloramphrnicol

- If still anorexic after 5 days, continue for 10 days

- If still, reassess checking for infection sites, resistant organisms and you may try Vit/mineral sup.

6- Correct Micronutrient DEFECIENCIES:

Vitamins, zinc, copper and iron

Vitamins: vit A at day 1 (same doses)

Folic acid (at day 1), Multivitamin complex

Iron: never in first week (increase infection), start after gaining wt (wk2) after stabilization phase

N.B: electrolyte / mineral solution contains: Zinc, copper, K, mg

7- START CAUTIOUS FEEDING:

By F-75 (milk based)

Components F-75 (100ml) contains:

-Calories 75 kcal

-Proteins 0.9 g

- Use day 1 wt to calculate how much to give

HOW TO GIVE:

Day 1-2	2 hrly	11ml/kg
Day 3-5	3hrly	16 ml/kg
Day 6-7 +	4 hrly	22ml /kg

How to give? by cup (no bottle)

NGT if: eat <80% (< 105ml/kg / day) or

Feeding problem (cleft palate)

Monitor:

1- Amount of food offered and left over

2- Diarrhea/ vomiting

3-Daily body wt

8- ACHIEVE CATCH -UP GROWTH (rehabilitation phase):

By **F-100** or **RUTF**

Components: (100ml of F-100)

100 Kcal, 2.9gprotiens

Difference from F-75:

F75: high sugars, low vegetables, low milk, no creal flour in F-100

When to start F-100: significant rehabilitation, regaining of apitite , usually after one week ,decreased edema .

Transition from F-75 to F-100 is gradual to avoid HF .

Transition: if the child eats < 50 % of the amount, back to F-75

You enter if he eats > 75% of F-100 or RUFT

-If during feeding HR increased by 25 and RR increase by 5 or more>>> lower the dose

-Aim of rehabilitation phase: rapid wt gain > 10 g /kg /day.

- Check if feeding target is met;

< 5-10 g/kg/day → look for infection

<5g/kg /day is poor need full assessment

10- PREPARE FOR FOLLOW UP AFTER RACOVERY:

Recovery: 90% wt for length (-1SD)

***CRITERIA FOR CURE by time*:**

1- 3-4 days→disappearing of apathy and irritability
2- 10days →skin lesions heal
3- 10 wks→child loss wt then gain wt 2.3kg
4- 3 Months →Return to normal health
5- 5-albumin > 3gm dl

Some ER

SHOCK	***ANEMIA***
The cause is either: - Dehydration or Septic shock ***Treatment:*** - Oxygen -I.V 10% D -Abx ✓ I.V fluids (15 mg /kg /hr over 1h) 5%D + ½ NS / R.L If not all → R.L -If child is improving (↓PR, ↓RR) : -Repeat fluid dose - Start ReSoMal(dehydration) -If no improvement ,give maintenance fluid(4ml/kg/hr) while waiting for - Fresh whole blood (10mg /kg over 3 hrs) **When to stop blood transfusion?** - If transmission reaction occur (fever , itchy rash, red urine, shock ,confusion) -RR increase by 5,HR increase by 25	Mild /moderate: iron in rehabilitation for 2 months -If Hb< 4 or PCV <12% or 4-6 times respiratory distress →blood transfusion as above or packed cell (5-7), if he has signs of CHF (do not transfuse within 4 days) ***N.B*** : TB , if strongly suspected (Hx of contact with TB pt ,poor growth despite good intake ,chronic cough ,chest infection not responding to Abx),do: -Monteux test - Chest x-ray. If test is +ve or strong suspicious , start anti-TB

*RICKETS:

Definition: Defective mineralization of bone

Causes

Nutritional (infantile)	- Breast feed only after 6 months - No sun exposure - Malabsorption
Impaired metabolism	- CRF - Chronic liver disease - Anticonvulsants (e.g. phenytoin)
Others	-Familial hypophosphatimicrickts (Vit D resistant rickets),no phosphate : no mineralization . or Non familial, commonest cause is Fanconi anemia -Vit D dependent rickets: **Type 1**: no 1α hydroxylase **Type 2:** unresponsive receptors.

C/O: **Genera**l: - Failure to thrive - Short stature *Square box shape

* **BONES:**

- **Head**: ↑H.C, frontal + parietal bossing + flat occipital, large unclosed fontanels, cranio tapes (earliest change)

N.B hydrocephalus, osteogenesis imperfecta: DDx for craniotapes just not all head features

- **Thorax:** - Rachitic rosary (costochondral junction)

- Harrison sulcus (transverse)

- Longitudinal sulcus (pigeon chest)

- **Extremities**: bowing, #

- **Pelvis:** narrowing

- **Muscles+pott-pelly abdomen**: weakness + hypotonia→Repeated chest infection also because of immunodeficiency

Delayed milestones

- Convulsions due to hypocalcaemia.

****Lab: Bone profile**: ↓Ca+2 , ↓ PO_4 (in familial ↓↓↓), ALP: ↑, PTH: ↑ (except in familial)

- ALP →normal in hypoparathrodisim , it is zinc dependent so is not elevated in malnutrition

Vit D:

	25(OH) D3 (liver)	1,25 (OH) (Kidney)
Type 1 Dependent rickets	–	↓
Nutritional	↓	↓
Type 2 Dependent rickets	–	↑

**Familial (x-linked dominant), males (short stature)

** Type 2 Dependent rickets (AR) (Alopecia)

Management:

- ❖ Nutritional rickets:
1. Vit D Daily 3000-5000 (changes in x-ray 6-8 wks)
2. OR single dose (IM/oral) 600,000 (changes in x-ray by 1 month)
3. Oral Ca+2
   Monitor by ALP ↓/ Vit D level / x-ray: healing
   Duration: 3 months
- ❖ Hypophosphatimic→ oral PO_4 sup.
- ❖ Vit D Dependent rickets →Vit D (in type 2 higher dose) , oral Ca
  OR Ca infusion to bypass intestine for 1 year

Wrist x-ray in rickets:

1-Clubbing, fraying of metaphysis (broad)

2-↑ joint space

3-#, deformities

4-Wide epiphysis plate. When healing → lines of calcification

يدبر الأمر.. ضعها ملء وجدانك وانهض على ثقةٍ منها إلى شأنك

PART 3: NEUROPEDIATRICS

1. **Seizures**
2. **Epilepsy**
3. **CP**
4. **Peripheral motor disorders**
5. **Peripheral nerve disorders**
6. **Muscular dystrophies**
7. **Floppy infant**
8. **Ataxis**
9. **Stroke**

Seizures **Causes:**

1. Epilepsy (un provoked)	2. Non epileptic (provoked)
-Idiopathic (70% _ 80%) -Secondary(HIE ,congenital malformation) -Neurocutanous syndrome -Tumours.	-Febrile convulsions -Metabolic (decreased glucose ,increased or decreased Na ,decreased Ca , decreased Mg) -Infections (meningitis, encephalitis). -Trauma -Poisons , toxins

Febrile convulsion: Seizure + fever in the absence of intracranial infection cause rapid increase in temp. (Viral infection) - Incidence: 3% - Age: 6 months _ 5 years

- Genetic predisposition 10%

Types:

Simple	Complex
-Brief -Generalized tonic clonic -Once/ day -No focal neurological deficit	-Prolonged more than 15 mins -Focal -More than one / day.

*****Increase likelihood of recurrence in the same illness:

1. Younger age.
2. Low temp. At the start.
3. Short period b/w illness will lead to convulsions
4. Family Hx.

-Risk of epilepsy: 1% as general population

But if complex: 4% _ 12%

Management:
-Rx of seizures: ABC + blood glucose
If more than 5 mins: rectal diazepam 0.5 mg/kg
Or buccal midazolam (same dose)
Then wait for 5 mins if not stop manage as status.
-Rx of fever: reduce clothes, fan, paracetamol, adequate fluid intake while looking for the cause of fever.

*Admission, if:

1. First attack
2. In subsequent age less than 18 months (Complex seizure)
3. Social factors

N.B meningitis is not apparent in ages less than 18 months. (Start empirical antibiotics)
L.P is C.I if, 1. Unconscious 2. CV instability

Status: more than 30 mins or multiple without regaining consciousness.
Rx: ABC + blood glucose

1. If there is I.V access, apply I.V **lorazepam** 0.1 mg/kg or I.V **diazepam** 0.3 mg/kg.

-If there is no I.V access **rectal diazepam** 0.5 mg/kg or **buccal midazolam** 0.5 mg/kg.
Wait for 5 mins, if not stopped then

2. Repeat above drugs and wait for another 5 mins. if not stopped then
3. I.V infusion of **phenytoin** 18 mg/Kg with N.S 0.9%

-Or loading dose of **phenoparb** 20 mg/kg orally or by NGT.
N.B If on phenytoin don't give phenytoin
If not stopped

4. ICU : Intubation + general anaesthesia (**thiopental** 4mg/kg (I.U/ I.O))

Midazolam loading dose then 2 mg/kg/min maintenance dose.

Paroxysmal disorders (funny turns)
-there is broad D.D for children with paroxysmal disorders depending on thorough history and examination+ investigations.

Causes of funny turns

1-Reflex _ anoxic seizures (Palloid B h attacks)	Toddler + pain (head trauma) or fear / cold food / fever" 3fs">>>>>> become pale (a systole) + LOC (exaggerated vagal cardiac reflex).
2-Breath_ holding attacks "cyanotic B.H.A"	Infant or toddler + crying (temper) = blue i.e.; holds his breath.
3-Syncope.	-
4-Benign paroxysmal vertigo.	Recurrent episodes of vertigo associated with nystagmus>>> viral labrynthitis
5-Basilar migraine.	-
6-Others, fabricated, pseudo seizures, etc.	-

Childhood Epilepsy **Some epilepsy syndromes:**

WEST syndrome	4-6 months, infantile spasm on walking (**salaam spasm**→flexion then extension.) **EEG** :hypsarrythmia"chaotic pattern" **Rx:** ACTH. If caused by T.sclerosis>>vigabatran(visual field defect) **Complications**: epilepsy , learning difficulty ***N.B*** *Infantile spasm +hypo arrhythmia =WEST syndrome* ***DDx*** *: T.S* *Hypoglycaemia.*
LENNOX –GASTAUT syndrome	Multiple seizures types "1-3 yrs" (Atypical absence, tonic, a static "drop" attacks). Neurodevelopment arrest or regression and behavioural disorders
JUVENILE MYOCLONIC EPILEPSY	At adolescence " when absence end " Throwing foods in the morning. Lifelong Tx, remission is unlikely. Learning is unimpaired

Anti-epileptic drugs:

First line: Sodium valporate, Carbamezapine.

Second line: Lamotrigine, Topiramate.

Cerebral palsy:

Def.: Abnormal movement / posture caused by non progr essive damage to the growing infant or foetal brain (Damage before 2 yrs).

- Damage is non- progressive. But C/O Progressive. Incidence 2/1000 live births

Causes:

1. Antenatal (80%) , congenital infections , c.malformation , vascular problem, etc.
2. Natal (10%), HIE, Trauma.
3. Postnatal (10%), meningitis, encephalitis, hypoglycaemia, cardiopulmonary arrest, haemorrhage, etc

C/O "clues"

1. Abnormal posture (limbs /trunk).
2. Delayed motor milestones.
3. Asymmetric hand function (less than 12 months)
4. Persistent primitive reflexes.
5. Gait problems when walking
6. Feeding difficulties (gagging , vomiting , in coordination)

Types:

1. **Spastic 90%.**
2. **Dyskinetic 6%.**
3. **Ataxic 4%.**
4. **Mixed.(chorioathetoid)**

spastic	**Dyskienetic**	**ataxic**
Damage :corticospinal pathways *__Hemiplegic:__ arms more than legs, sparing the face. Cause: neonatal stroke. Arms: fist, flexed and pronated. Legs: tip toeing. *__Quadriplegic__ : Cause: HIE Most severe form and the most associated with learning difficulties and epilepsy. *__Diplegic__: Cause: PVL (preterm) **N.B;** Spasticity: crossing, tip toe gait.	Old , kernicterus now, HIE Damage : basal ganglia	Genetic Damage: cerebellum and its connections

***Problems of C.P or complications**

1. Learning difficulties
2. Epilepsy
3. Hearing loss
4. Visual impairment , squint
5. Speech problems
6. Feeding problem>> growth failure.
7. Joint contractures.

Rx is multidisciplinary -Physiotherapist, orthopaedic surgeon, speech therapist, adequate feeding, etc.

CP



PERIPHERAL MOTOR DISORDERS:

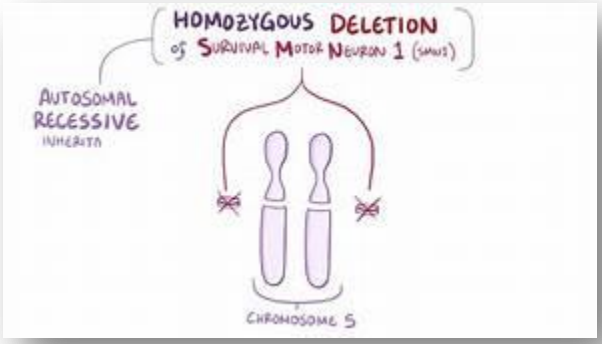
- Degeneration of motor neurons in the anterior horn. (Fasciculation weakness)

**Spinal muscular atrophy SMA:

-AR "survival motor neuron (SMN) gene

Type 1	Type 2	Type 3
(Warding - Hoffman's) C/o: -foetus with decreased movement -At birth :arthrogryposis (deformity + contracture of at least 2 joints) then LMN features "remember tongue fasciculation's" -Never sit unaided , death by respiratory failure at 12 month "diaphragm paralysis" -Gold standard Dx is genetic study.	(intermediate): sit but never walk independently	(kugelberg-Welander): Later in life (can walk).

**Polio: Asymmetrical flaccid paralysis



Peripheral nerves (areflexia/ pescavus/ distal weakness):-

Hereditary motor sensory neuropathy (HMSN):

TYPE 1 (Charcot -Marie – Tooth disease)

-Peripheral muscular atrophy

N.B inverted wine bottle leg.
-AD. 1st decade: distal weakness/ wasting and pescavus.
DDx: F. Ataxia
-Normal life span. Nerves: enlarged "remyelination attempts"
Gold standard Dx: sural nerve biopsy. Biopsy: onion – skin.





Fasciculation



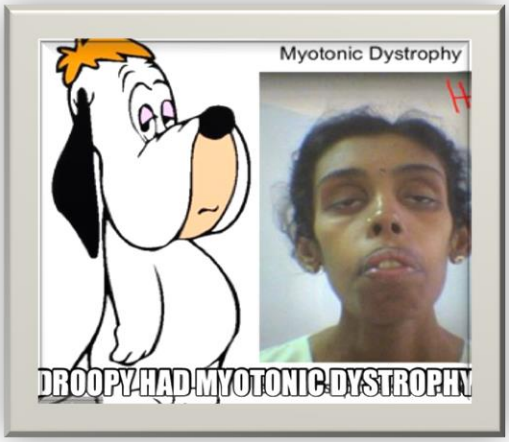
Muscles: **Muscular dystrophies:-**

Duchene	Becker's
X-linked recessive (dystrophin) C/o proximal weakness >>>waddling gait, gower sign. Age 4 - 5 years. Increased CPK. At 10-14 yrs : not ambulant Late 20s :death "respiratory failure , cardiomyopathy DCM" N.B Scoliosis is common. Rx 1. Physiotherapy 2. Respiratory: NIV at night.	Same but slower Dx: 11years "Duchene 4 to 5 yrs" Unambulant : 20s "Duchene 10 to 14 yrs" Death : late 40s " Duchene late 20s"

****Myotonia:**-Delayed relaxation after sustained contraction.

<u>MyotoniaDystrophica:</u>** - AD (trineucleotide repeat) -Association: frontal bladness, testicular atrophy, cataract, learning difficulties and DM.

Death: cardiomyopathy.



<u>Neuromuscular disease: illustrated</u>
p.480

<u>FLOPPY INFANT:</u>

Rag – doll, head lag, slip through fingers
. Frog – like posture (abduction of LLs)

Causes:

Central	1. **Cortex**: HIE, C.malformation. 2. **Genetics:** Down's, prder-willi. 3. **Metabolic**: hypothyroidsm, hypocalcaemia.
Peripheral	1. SMA 2. Myopathy 3. Myotonia 4. Congenital myasthenia



ATAXIAS: **Both are AR**

1. Fr.ataxia	2. Ataxia telangectasia
Gene:frataxin *Degeneration : -Spinocerebellar→ataxia Corticospinal→weakness + up going plantar reflex -P.nerves→areflexia (all reflexes) &wasting. -DCT→Loss of vibration &proprioception. -There is pescavus "high arch "so DDx is HMSN. -Death 40 – 50 (HOCM, kyphosciliosis)	DNA repair gene (ATM) -Infancy: oculomotor dyspraxia. -4 yrs: telangectasia (eyes, shoulders, neck). -School age: ataxia. "sinopulmonary" -Associations >> IgA deficiency = infections. -Increase risk of ALL -Increase alpha fetoprotein Sensitivity of WBCs to radiation "remember Fanconi"

Other causes of ataxia:

- Infections: e.g. varicella
- Drugs: e.g. .phenytoin
- Posterior fossa neoplasm e.g. medulloblastoma

Telangectasia



Stroke: **Causes:**

1. Cardiac: cyanotic CHD
2. Heamatologic:SCD,anti-thrombin def
3. Post infective: varicella
4. Inflammatory:SLE
5. Metabolic\genetic :homocystinuria
6. Vascular malformations: moya-moya disease(abnormal vasculature).

Investigations: - MRI, MRA, carotid Doppler studies - Echo - Metabolic test.

Child may need rehabilitation Aspirin prophylaxis.

Neural tube defects

-Failure of neural plate fusion to form neural tube in the first 28 days of life.

It include:

Anencephaly :	***Encephalocele***	***Spina bifida occulta***	***Meningiocele***	***Myelomeningiocele***
Rx TOP (termination of pregnancy). Most severe: dead "still birth" or die shortly after birth	Neurosurgery , but usually there is underlying brain malformation	Failure of fusion of v.arch Skin : tuft of hair , birth mark , lipoma , sinus Next step >>> U/S or MRI WHY? To exclude tethering of the cord "diastomatomyelia" If +ve >>>neurosurgery.	Neurosurgery ASAP: Risk of meningitis. Good prognosis.	C/o 1-LL paralysis→ (walking aid , physiotherapy) 2-Muscle imbalance : dislocation , talipes→(physiotherapy) 3-Loss of sensation→ (skin care) 4-Neuropathic bladder →(intermittent catheterization, oxybutinin be ware of UTIs , RF , hydronehrosis) 5-Neuropathic bowel→ (regular toileting , laxatives) 6-Scoliosis→ (surgery) Hydrocephalus 80% due to Arnold Chiari malformation→ (shunt) Most severe disease : above L3

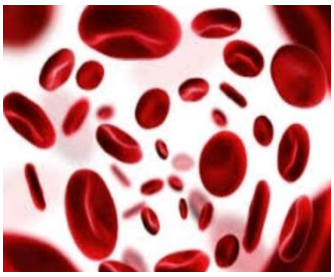
Hydrocephalus الشيت **N.C syndromes illustrated page 489**

حتى تنجح ... لا تعيش حياتك وأنت ترسم الحدود

عش حياتك وأنت تتخطى كل الحدود

PART 4: HEMATOLOGY

1. **Anemias**
2. **Sickle cell disease**
3. **Thalassemia**
4. **Anemia in newborn**
5. **Coagulation problems**



Anemia: Main site of hemopoiesis in the fetus => liver.

	Hb F	Hb A	Hb A2
Fetus	100%	-	-
At birth	75%	26%	1%
After one year	-	97%	2%

***Differences b/w Hb F /Hb A: Hb F:** higher affinity for O2.Dissociation curve: shifted to the left.

At birth:

- Hb is very high (14-21g), gradually decrease over two months (10) => lowest value .
- Blood volume: term 80 ml /kg , preterm 100 mg/kg
- WBCs = higher (10-25)
- PLTs = as adult.
- Iron, B12, folic acid => adequate stores in both term and preterm.

But in preterm: more ↓↓ and its lower (iron, folic acid).

***Diamond black fan anemia:**

-Family hx =20% , Sporadic= 80%

-C/O: At birth (25%) , rest in 2-3 months.

Maybe associated with short stature, abnormal thumb .

-Gene RPS (ribosomal protein).

-Tx: Steroids. If unresponsive, monthly RBCs transfusion, stem cell transplantation.

***Transient erythroblastopenia:**- No **RBS gene** - Resolves in few weeks.

- No family hx, no RPS, congenital anomalies.

Triggered by viral infection

Increased RBCs destruction (hemolytic anemia) :-

↑Ritcs, ↑Bilirubin, ↑urobilinogen Hepatosplenomegaly.

Some points

*G6PD deficiency = most common enzymepathy.

= most common cause of severe neonatal jaundice requiring transfusion.

- Hemolysis in G6PD deficiency => dark urine, fever.

Sickle cell disease:

Autosomal recessive inherited hemglobinopathy.

1- **HbSS** (SC anemia): no HbA
2- **Hb SC:** no HbA
3- **Hb SB:** Thalassemia no HbA
4- **Hb SA (**SC trait):40% of Hb is Hb S

Sickle cell crisis	Dx	Mx	complications
1- vaso-occlusive crisis 2-hemolytic crisis 3-aplastic crisis 4- splenic sequestration crisis	1-full blood count 2-blood film 3-sickling solubility test 4-Hb electrophoresis.	**Prophylactic** : 1- full immunization 2-daily oral penicillin 3-folic acid 4-hydroxy urea 5- Avoid precipitants (warm , adequate hydration before exercise **Mx of acute crisis:** 1- hydration 2-analgesia 3-oxygen 4-Abx 5- blood transfusion if needed	1-Growth failure (delayed puberty). 2-HF --> anemia. 3-Gall stones. 4-Renal impairment => exacerbate enuresis (inability to concentrate urine). 5- Leg ulcers. (Uncommon in children). 6- Psychological problems. 7-stroke

Indications of exchange transfusion:

- Sequestration crisis
- Acute chest syndrome
- Stroke , priapism

Thalassemia: Defects in production of the α or β chains of hemoglobin resulting imbalance in globin chains leads to ineffective erythropoiesis and Hemolysis in the spleen or BM

Types of Thalassemia:

- **β.Thalassemia** :-

-β- Thalassemia major => no Hb A

-β- Thalassemia trait (intermediate) => few Hb A , high Hb f .

β- Thalassemia major:

C\O**:-** severe anemia (Transfusion – dependent)

If no transfusion =>

- Extramedullary hemopoiesis(frontal bossing, maxillary overgrowth)
- Growth failure.

Mx:- Regular blood transfusion + iron chelation (oral \ SC desofrroxamine) **aim: Hb = 10 **

Complication of blood transfusion:-

1. Hemochromatosis (cardiomyopathy, growth failure …etc).
2. Abs formation (10%).
3. Infections (Hep B, C \ HIV \ malaria).
4. Venous access problems.

** Definitive => BM transplantation**

β- Thalassemia trait:

Sx: HbA= >90%

DDx: IDA. (Check ferritin)

N.B* Dis ↓ in MCV, MCH "very low", RBCs may be even↑.

-Most important in Dx➔↑Hb A2.

- **α.Thalassemia** :-4 genes ,

  All absent➔Hb Part (hydropsfetalis)

  3 absent ➔Hb H (Moderate anemia)

  2/1 absent ➔α.Thalassemia trait (mild/no anemia)

  **NB; β chain X ➔ after 6 months, while α chain X at birth

  ** Perinatal diagnosis ➔ chorionic villous sampling.

Thalassemic face



Anemia in Newborn:- ***Causes***

↓ Production	congenital B19 infection, Diamond black fan
↑Destruction	Immune / non immune(G6PD def., HS, α.Thalassemia)
Blood loss	To mother (fetomaternalhemo.) / to twin (TTTTS) / at delivery ** ↑ Ritcs, normal Bilirubin
Anemia of prematurity	↓RBCs life span, inadequate erythropoiesis, ↓iron& folic acid, frequent blood sampling

Bone marrow failure (A plastic anemia): **Pancytopenia

- Inherited: Fanconi, shawrman's diamond **- Idiopathic**

- Acquired: Infections (hepatitis), Drugs (sulfa amide, chemotherapy), toxins (benzene)

Fanconi anemia	***Shwachman Diamond Syndrome (SDS)***
AR (DNA repair gene) ***C/O:*** • At birth: congenital anomalies (absent thumb),macropthalmia, Kidneys (horse-shoe, absent), skin (café aulait) • 5-6 years: Pancytopenia ***Dx:*** When exposed to damaging agent → ↑ chromosomal breakage of peripheral blood lymphocytes Prenatal Dx, family screening ** ↑ risk of acute leukemia, other malignancies (e.g. GI, gyne) ***Mx:*** B.M transplantation	- AR. - Association → Exocrine pancreatic abnormalities, skeletal abnormalities, also increase risk of leukemia. ***Dx:*** -prenatal Dx. - Genetic study (SBDS).

Coagulation problems:

Inherited	Acquired
Hemophilia -VWD	-V k deficiency -liver diseases -ITB -DIC

PT = extrinsic pathway. APTT = intrinsic pathway. PT = not used now. (Platelet function analyzer). RiCoF "Ristocitin cofactor" = Vwf activity.(↓Vit D)

Ristocitin- induced platelet aggregation = abnormal in VWD.

Hemophilia	VWD
- ↑APTT. - X-linked / recessive. - A= factor VIII deficiency. B= factor IX deficiency. C/O: Neonate ➔ I.C hemorrhage, post-circumcision/ venipuncture bleeding. Toddlers➔ when start to walk / crawl. ***Severity**: - < 1% severe→spontaneous bleeding - 1-5% moderate→ bleed after Minor injury. - 5-40% mild →after major injury. **Mx:** Very mild bleeding = DDAVP (desmopressin: synthetic ADH). Others: factor VIII concentrate How much? If minor bleeding: till conc. reaches 30%. If life- threatening: till " " 100%. Then maintain 30-50% for 2 wks. Severe hemophilia = F VIII prophylaxis. (↓Risk of joint damage). also= Hemophilia (including psychosociotherapy, physiotherapy). ** avoid: I.M injection / NSAIDs- aspirin	-AD ↑BT, ↑APTT. -VwF:→ platelet aggregation, factor VIII carrying and protection. -Many subtypes: # Subtype 1 = mild (commonest). **~ C/O** : puberty/ adolescence: Skin bruising. Mucosal bleeding (menorrhagia/ epistaxis). ****N.B**; Spontaneous bleeding not common. **Tx** : -DDAVP → mild (subtype 1). Risk: hyponatremia (=seizure) especially if < 1 yr. -F VIII concentrates (plasma – derived only). Not recombinant i.e.; do not containVwf. -replacement therapy. Also avoid: -Aspirin – NSAIDS. -I.M injections.

Complications of Hemophilia:

• **Joint bleeding**
– Knees>elbows >ankles >shoulders – Leads to chronic synovitis
• **Muscle bleeds**
– Leads to fibrosis and atrophy – Can lead to compartment syndrome
• **Pseudo tumors** • **Subdural hemorrhage**
• **Inhibitors** • **HIV**/hepatitis C – 90% if Rx started before 1994



Thrombocytopenia: Causes:

↑Destruction	Immune: ITP , SLE Non-immune: HUS , TTP , DIC , hypersplenism
↓ production	Congenital :wiskott Aldrich , Bernard Soulier Acquired: BM infiltration (leukemia, a plastic anemia) .

PLt: If 50 - 150.000 = mild (major surgery/trauma)

< 50 – 20 = moderate (moderate "surgery). < 20 = severe (risk of spontaneous bleeding)

Bleeding: bruising, petichea, mucosal, GI , intracranial (rare)

***ITP (immune thrombocytopenic purpura):**

- Commonest cause of thrombocytopenia in children. - IgG Abs against platelets.

C\O:-
- Previously health child - 1-10 years - Sudden onset of generalized purpura and petichea

- Profound thrombocytopenia - Bleeding from mucus membranes and gums.

#Dx: - by exclusion (ALL, SLE …etc)

N.B: you should examine the bone marrow if you want to give steroids, it will mask ALL.

#Mx:- Self limiting in 6-8 wks.

No TX unless:

1. Severe bleeding (I.C, GI) . 2. If minor but affecting life

➔ Oral prednisolone or IVIG or IV anti D Avoid contact sport when platelets are low.

****Chronic ITP** = > 6 months Also no TX when: Same indications

→Rituximab (monoclonal Abs against B lymphocytes).

- Splenectomy (non-responsive pt)



ITP

DIC: Sepsis, shock , major trauma , burns

Lab: ↑PT, ↑PTT.

- ↓PLTs, ↓fibrinogen. (Also ↓ protein C/S, ATIII).
- ↑f. degradation products , ↑ D-Dimers

Mx:

- Ttt of underlying cause.
- F.F.B or cryoprecipitate
- PLTs. (if needed)
- ATIII, protein C concentrates. (if needed)

كن راضيا وكأنك تملك كل شئ

فما كتبه الله لك ألطف مما تشاء

PART 5: RHEUMATOLOGY

1) **Juvenile idiopathic arthritis**
2) **SLE**
3) **Juvenile dermatomyocitis**



▪ Juvenile idiopathic arthritis:

The most common chronic rheumatological disease in children.

- Genetic susceptibility.
- Environmental triggers.

▫ **C/O:**

- Morning stiffness (better during the day).
- Arthritis or 2 of (joint pain / tenderness,
- ⬇ range of movement, worm).

▫ **Criteria**:

- Onset < 16 years.
- Joint swelling ≥ 6 weeks.
- No other causes.

▫ Long term: Overgrowth:

- Discrepancy in length, genu valgus
- Digits different in length.
- Wrist advance bone age

▪ Types of JIA

- Systemic arthritis
- Polyarthritis
- Pauciarthritis or oligoarthritis
- Entheistis-related arthritis.
- Psoriatic arthritis.

Pauciarticular	Polyarticular
50%, Commonest. **<5 joints.** **Usually large (knee, ankle, wrist) not hip** **F>M** **ANA +ve in 20% (⬆ risk of uveitis)** **Prognosis good.** ***N. B. If evolve to involved ≥5 joints after 6 months = extended Pauciarticular type.**	≥5 joints (symmetrical) Usually small joints. F>M If RF +ve "early onset RA" (F adolescent 10-14 years) worse prognosis than RF - ve. Cervical spine may be involved.

▫ **Systemic onset JIV**:

Fever, Hepatosplenomegaly, lymphadenopathy, salmon rash.

⬆ WBCs, ESR/CRP, Platelets, Anemia.

Worst type especially if Polyarticular.

▪ **Other types:**

Enthestitis- related arthritis	Psoriasis related arthritis
Inflammation of tendon area e. g Achilles' or arthritis + ≥2 of : o 1. uveitis o 2.spinal pain o 3.sacroilietis o 4. FamilyHx of HLA. B27 related diseases (IBD, uveitis, and spondyloarthritis). o 5. HLA. B27 M>F	Arthritis + psoriasis or Arthritis + family Hx of psoriasis + dactilitis, finger nail changes.

▪ **Complications:**

- Chronic A. Uveitis.
- Joint contracture = needs joint replacement
- Amyleidosis.
- Osteoporosis (nutrition, steroids, ⬇Wt bearing).
- Growth failure, delayed puberty.

▪ **Tx:** Multi disciplinary: ❊ **Drugs**:

1st choice: NSAIDs (not aspirin)

Avoid steroids unless severe sys. Onset JIA or severe arthritis (unable to walk).

2 nd line: - Methotrexate (hepatotoxicity, B.M suppression)

- Sulfasalozine, hydroxychloroquine...etc

If failed = anti TNFα (inftiximab, etanarcept)

* **N. B** if you do arthrocentisis: WBCs 50,000 – 100,000, lymphocytes (no neutrophils) → non suppurative

Remember: physiotherapy, ophthalmic follow u




▪SLE:

Criteria : MD SOAP N HAIR

M = Malar rash
D = discoid rash (scaling)
S = serositis...
O = oral ulcer
A= +ve ANA
P = photosensitivity
N = neurological (seizure, psychosis)
H = hematology (+ve comb Anemia, Thrombocytopenia, lymphopenia).
A = arthritis≥2 peripheral joints.
I = immunology (anti ds-DNA, anti Smith Abs, false +ve tests for syphilis, anti cardiolipin Abs= thrombaitis).
R = renal (proteinuria, cellular casts)

Other manifestations:

Reynaud's
Best for screening (highest sensitivity) = ANA
Specificity:

- Anti ds DNA. ⬆During active disease
- Anti Smith Abs. Most specific

How to follow:

- Anti ds DNA. Best for follow up.
- ESR = in RA, (CRP is normal).
- Complements: it is low first.

▪Tx:

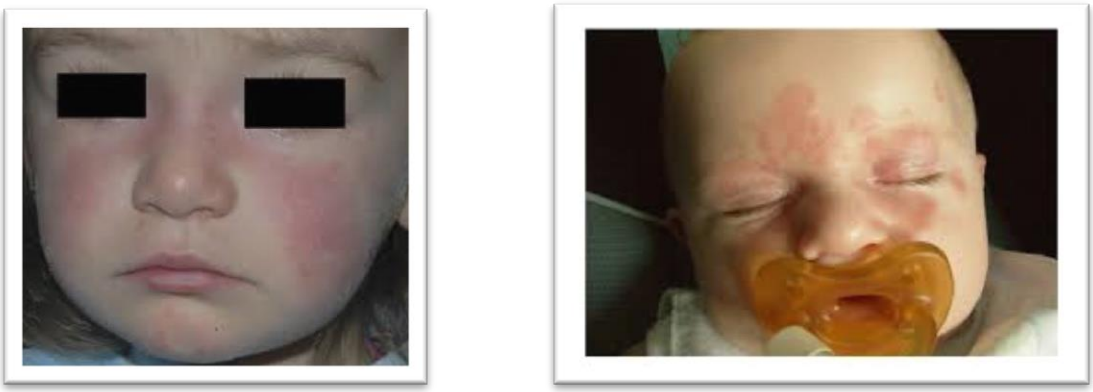
- Severe flares (nephritis, CNS, Pul.Hemorrhage. etc). →**I. V cyclophosphamide**.
- Maintenances + skin disease→**Hydroxychloroquine** (monitor eyes). - Steroids as start

▪ **Neonatal lupus:**

- Maternal Anti Ro. Anti LA, Abs.
- Rash, hepatitis, hematological→ reversible.
- Congenital Heart block→ Permanent.

TX: - Corticosteroids - IV.IG - Cardiac pacing

Severe cardiomyopathy>>cardiac transplantation.



Neonatal lupus

▪ **Juvenile Dermatomyocitis:**

- Most common idiopathic inflammatory myopathy in children.

- Auto immune capillary vasculopathy>> genetic susceptibility. - 4 _10 yrs. (F>M).

C/O:

- Symmetrical proximal weakness (Difficulty with stairs, getting up, +ve Gower's.. etc).
- May affect pharyngeal muscles =dysphagia. - May affect Resp. Muscles.
- Skin :

- Rash. (Face)
- Shawl's sign.
- Gottrons papules.
- Heliotrope Rash.
- Periungualr erythema.

- Arthritis

Dx:

- No sys : inflammation (just muscles), WBCs, ESR, CRP are normal
- EMG(electromyography changes) : myositis
- ⬆Muscle enzyme (AST, ALT, CPK, LDH, Aldolase).
- Biopsy : confirmatory

Mx:

- Steroids
- Skin: hydroxychloroquine , avoid sun(= SLE)
- #1 complication →Calcinosis

Heliotrope rash



Gottrons papules





Shawl sign

A predominance of rash over the neck and upper shoulders has been termed the ***V sign***, while rash over the shoulders and upper arms, the ***shawl sign***.

This distribution suggests that the skin changes reflect ***heightened photosensitivity***

The Pictures of Juvenile Dermatomyositis



Heliotrope Rash with periorbital edema



Gottron's Papules



- Malar rash
- Rash on extensor surfaces



V Sign



Shawl Sign



Calcinoses

لا تستسلم كلما تعثرت انهض وكلما اخطأت صحح وكلما فشلت حاول

وكلما اصرت الايام ان تجعلك عابسا ابتسم رغما عنها

PART 6: EXANTHEMAS

A- Maculopapular rashes

1- Measles
2- Rubella
3- Fifth disease
4- Rorseola infantum
5- Scarlet fever
6- Infectious mononucleosis

B- Vesicular rash

1- Varicella zoster
2- Hand foot and mouth disease
3- Kwasakis disease

Measles: Paramyxovirus * Infectivity: 4 days before- 4 days after the rash

C/O: High grade fever+3Cs (coryza, conjunctivitis, cough)

Koplik's spot (oral mucosa) →maculopapular rash (starts in the face→behind the ears→body)

Season: late winter, spring

Complications:

1- Respiratory: pneumonia, otitis media (most common), and tracheitis
2- CNS:

A-encephalitis	B-Sub acute sclerosing pan encephalitis=SSPE	C-Febrile convulsions
-7 days after starting illness -15%=mortality -40%of survivors=long term sequel(deafness, convulsions,)	-7 years after infection -Dementia death -very high measles Abs in the serum/CSF	-

3- Others: appendicitis, hepatitis,

Long term:

- Diarrhea→dehydration
- PEM (protein energy malnutrition)
- Corneal cloudication
- Mouth ulcers

***Measles in special conditions (low CMI):

PEM: severe disease HIV: modified or absent +rash →severe disease

Mx: - Antipyretics - Eye care=tetracycline ointment - Mouth care: salted water

- Nutrition + Vit. A - Ribavirin: immune compromised only





Rubella (German measles) Mild disease Risk= Congenital rubella syn.

C/O: prodrome: mild or no fever

Rash: maculopapular (face →trunk) not itchy

Disappears in 3-5 days (3 days measles)
Exanthem→forscheiner spot in soft palate

Dx: serology Most characteristically= Sub occipital post auricular lymphadenopathy

N.B: since fever is mild or absent the rash is usually the first sign

Complications:

- Congenital rubella syndrome - Arthritis in female usually - Myocarditis

Prevention: MMR



Erythema.infectiosum (Fifth disease)

- B19 disease_____As x

Prodrome (wk.) →slapped cheek appearance→

lace reticular rash in the trunk+extrimities

Recurrence can occur; proactive factors→

Heat (sun exposure, hot path, emotional upset



Complications:

- Arthritis (adults) - A plastic crises→

1/chronic Hemolysis

2/immunodeficiency (can't form Abs to clear infections) E.g. malignancy

- Non immune hydropsfetalis (vertical transmission); If affects pregnant women in the first trimester

N.B: B19 affects eryothoblastoid cells (precursors of RBCs) (Bone marrow)

<u>Roseola infantum (Exanthemsubitum)</u>

- HHV 6, HHV 7 - Infant, high fever (but he is relatively well

- Fever disappears in 3or4 days

- Then rash appears in hours (rarely 24 hrs.)

- Starts from the trunk→extremities→neck→face

N.B: miss diagnosed as drug allergy

Infant +fever=Abx, rash appears



<u>Scarlet fever</u>

Only rash covered by bacteria Group A streptococcus

C/O:Fever, exudative pharyngitis, tonsillitis - Abdominal pain, DDx for acute appendicitis

- Maculopapular rash (goose like skin or sand paper) especially deep areas= antecubital fossa, axilla, groin), areas of hyper pigmentation that doesn't disappear by pressure

- White coated tongue then strawberry tongue

- Desquamation in 6 days sunburn like skin (esp. in fingers/ toes) +circumoral pallor.

TX: Penicillin

Complications:

1. RF
2. Peritonsillar abscess (hot potato like voice)
3. AGN
4. Retropharyngeal abscess



<u>Infectious mononucleosis (glandular fever)</u> -EBV

Association with EBV (oncogenic) 1/Burkett's lymphoma 2/Nasopharyngeal CA

Transmission: oral secretions (kissing disease)

C/O: fever, tonsillopharyngitis (may obstruct breathing)

Lymphadenopathy (prominent cx) large Splenomegaly, hepatomegaly, rash

N.B: if treated with amoxicillin→amoxicillin rash

Dx: Atypical lymphocytes (peripheral blood) Heterophil Abs +ve (monospot test is +ve)

Serology: high EBV IgM, IgG

TX: supportive, avoid contact sporting (spleen rupture)

Vesicular rash

Varicella zoster (chicken pox)

Infectious: 2 Days before the rash→until crusting of all lesions (5-7 days)

Rash: itchy, vesicular, starts in the trunk in different stages (macule, papule, pustule, vesicle, crust) Goes in crop Sparing palms and soles

Complications:
1/2ry bacterial infection s.aurus, strep (most common complication)
-Mild or sever as toxic shock syndrome, necrotizing fasciitis
How to suspect? Recurrence of fever
2/CNS: encephalitis (better prognosis than HSV), cerebellitis
3/ in immune compromised: pneumonitis hemorrhagic vesicle, disseminated disease (may be DIC)
4/Purpurafulminans=thrombocytopenia

TX: Supportive - Antiviral: IV acyclovir or oral Val acyclovir→Immunocompromized

Oral Val acyclovir=adults

VZIG → Immune compromised in contact with diseased person

Severe disease:
adults+immunocompromised





Hand foot and mouth disease:

- Vesicles in these areas +buttocks

- **Cause:** coxackie virus, A16

Dorsum of the hands and feet, may affect palms and soles

Kawasaki disease(mucocutaneus LN disease)

- Vasculities - Target: 6months-4years (<5)

C/O: Fever >5 days + 4 of:

1/non purulent conj. Injection

2/mucus membranes (red, cracked lips, pharyngeal injection, strawberry tongue)

3/rash: polymorphous, no vesicles, no crust

4/cervical lymphadenopathy (1 L.N >1.5 cm)

5/extremities changes (palms soles edema, peeling (conuelsence), erythema

Lab: high WBCS/ESR/CRP/platelets (2nd wk.)

Complications: 1/ Coronary aneurysm (6wks) 2/ MI (thrombosis) 3/ Sudden death

TX: -IVIG (within 10 days) +high dose of aspirin until fever, inflammatory markers are down

Then change to low dose aspirin (prevent thrombosis)

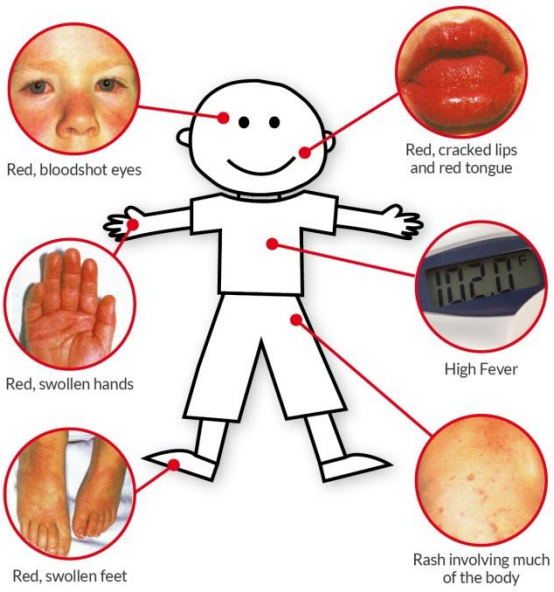
Echo: If +ve aneurysm= warfarin+ follow up

N.B: Aspirin is given for 6 wks. First high dose=anti-inflammatory. Then low dose=ant platelet

If poor response to aspirin; infliximab or steroids

N.B: IVIG within 10 days→decrease incidence of coronary aneurysm





فصبرا في مجال العلم صبرا فما نيل المنى سهل المراد

PART 7: IMMUNIZATION

Passive	Active
- Natural: from the mother - Acquired: Igs or antitoxins	- Natural: post infection - Acquired:vaccination

- May be life long as in MMR & chicken pox
- EPI: Eradication of poliomyelitis, measles & Neonatal Tetanus

N. B.: 10 childhood infections + T. Toxoid

- ***Types of vaccines:***
  1. Live attenuated vaccine : most potent (BCG, OPV)
  2. Killed vaccine
  3. Recombinant vaccine : Hep B
  4. Capsular polysaccharides vaccine : pneumococus&Hib

Contraindications to all vaccines:

1. Previous history of allergy (edema, breathing difficulties, shock)
2. Severe local reaction
3. Fever >40 c° or severe illness.
4. Within 6m of blood transfusion

For all live attenuated:

- Pregnancy - Immunocompromized with exception of MMR & varicella in child with HIV infection

Schedule:

Day 0	- BCG(0.05)ml ,left upper third of forearm, intradermal - OPV(zero dose (Sabin))2 drops oral
6 wks	- OPV 1 - Pentavalent: 0.5 ml. IM. Left Anterolateral thigh - Rota: 1ml oral. Live attenuated - Pneumococcal vaccine: 0.5 ml IM/SC Right Anterolateral thigh. Capsular conjugated to protein “Diphtheria Toxoid”
10 wks	- OPV 2 - Rota - Pentavalent - Pneumococcal
14 wks	- 0PV 3 - Pentavalent - Pneumococcal
9 months	- Measles : 0.5 ml. SC. Live attenuated
18 months	- Measles

New: IPV "Salk" gives in 3 doses with Pentavalent

N.B. Pneumococcal vaccine:

- 13 valent " immunization "
- 23 valent (SCD, Splenectomy, Nephrotic syndrome, Immunocompromized, cochlear implant, chronic diseases)

N.B

- MMR = Autism , IBD
- Rota = intesseception
- Pertussis = Encephalopathy

Measles = Thrombocytopenia

Side effects:

- All.: Mild local reaction (redness, swelling), Mild symptoms (e.g. Fever)
- BCG: lymphadenitis, skin ulcer, scar, local abscess.

N. B. Mild disease, low grade fever, vomiting, mild diarrhea =not contraindications

N. B.

Polysaccharides = B cells response > poor immunity less than 2 years, short term immunity, Absence of Booster response when exposure.

VS

conjugated polysaccharide = T cell response >⬆⬆⬆ level of functional Abs, long term persistent IgG response, immunological memory.

Special contraindications:

- BCG : prematurity, skin infection
- DPT(Pertussis) : Encephalopathy within 7 days, seizure within 72 hours
- Measles: Thrombocytopenia purpura, neomycin allergy, Egg allergy (caution), active TB.

Rota: gastroenteritis, hx of intusseption.

Dr. Mousab Mohammed Ibrahim batch 88 ((kibreet))

&

Dr. Emeirii Hatim Mustafa Elawad batch 88

إن ما يهون علينا دائماً أننا سنتذكر لاحقاً هذه اللحظة الثقيلة

التي تمر من فوقنا ونتنهد تنهيدة المنتصر

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